

Aspect-Based Sentiment Analysis (ABSA) Studio

Technical Design Report & Dataset Evaluation Benchmark

1. Executive Summary

Aspect-Based Sentiment Analysis (ABSA) is a granular Natural Language Processing (NLP) task that identifies specific entity features (aspects) in customer feedback and determines the distinct sentiment associated with each feature. While standard sentiment classifiers output a single coarse score for an entire document, ABSA Studio pinpoints actionable insights—such as separating praise for food quality from complaints about wait times.

2. Technical Engine & Rule Processing

The ABSA Studio core engine relies on a multi-stage rule-based extraction pipeline:

- **Aspect Keyword Matcher:** Scans tokenized sentences against domain-specific taxonomies.
- **Modifier & Negation Resolver:** Inspects contextual windows (+/- 3 tokens) around aspect targets to detect negations ('not', 'never') and intensity amplifiers ('extremely', 'slightly').
- **Clause Segmentation:** Splits compound sentences along contrastive conjunctions ('but', 'however') to prevent sentiment bleed across clauses.

3. Dataset Sample Results Summary

Domain	Aspect	Sample Term	Polarity Score
Restaurant	Food	Pizza (delicious)	+0.92 (Positive)
Restaurant	Staff	Waiter (rude)	-0.88 (Negative)
Hotel	Room	Spotless & spacious	+0.96 (Positive)
Tech	Battery	Drains in 3 hrs	-0.90 (Negative)
E-Commerce	Shipping	Arrived 2 days early	+0.94 (Positive)

4. Conclusion & Deployment

ABSA Studio provides a lightweight, instant, zero-latency sentiment extraction Web Application. It operates seamlessly as a standalone client-side or server-side application using standard Node.js tooling.